A knowledge-based early-warning system using artificial intelligence on venture data to predict digital-transformation failure in entrepreneurial firms

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Abstract

A warning that an entrepreneurial firm will fail a digital transformation has to name the mechanism, not only rank the firm. Public venture files record closure and funding, not that failure. This Short Communication presents KBEWS, a knowledge-based early-warning system that turns seven failure mechanisms into graded rules, combines the rules by certainty factors, and anchors the resulting score to a gradient-boosting probability. The output is a calibrated probability and the rules that fired. Across 80 independent synthetic cohorts of 500 ventures, mean AUPRC is 0.683 and the mean Brier score is 0.226. Paired Wilcoxon tests against gradient boosting give $p = 7.74 \times 10^{-13}$ for AUPRC and $p = 8.15 \times 10^{-15}$ for the Brier score. Against the knowledge base alone, ranking does not improve (AUPRC 0.683 versus 0.681, $p = 0.385$), while calibration does (Brier score 0.226 versus 0.270, $p = 7.85 \times 10^{-15}$). The claim is limited to this mechanism-labeled benchmark.

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No field failure rate is estimated.

Keywords: early warning, knowledge-based system, artificial intelligence, digital transformation, entrepreneurship, innovation, venture data

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1. Introduction

Digital transformation changes how a firm creates value, and the term now covers an organizational change rather than a software installation [1, 2]. Multidisciplinary reviews describe that change as a staged build of capabilities, strategy, and customer value [3, 4]. For an entrepreneurial firm the same technologies also change how opportunities are formed [5]. Ambition is high and slack is thin, so a miss is costly. A useful warning therefore has two outputs. It must score the risk of failure, and it must show which mechanism is active so a person can act on the score. Bibliometric reviews already treat digital-transformation failure as a research problem [6]. They do not supply a venture-level label a model can be trained on.

The mechanisms a warning would need are named in the management literature. Dynamic capabilities are an ongoing renewal process [7] and a condition of transformation success in industrial and shipping settings [8, 9]. In smaller firms, sensing and learning capabilities are what allow digital tools to change the business model and the offer to customers [10]. Leadership coherence, and a manager's digital literacy, are associated with whether that change advances [11, 12]. Dynamic managerial capabilities are studied as antecedents of digital business-model transformation [13]. An imbalance be-

tween exploration and exploitation remains the ambidexterity problem [14]. Analytics deployments yield sensing, seizing, and transforming capabilities only when organizational inertia does not block them [15]. Entrepreneurial transformation has been traced as information exchange with the operating environment [16], and digital entrepreneurship is also described as an ecosystem built from technologies and collective intelligence [17]. In a survey of smaller firms, digital orientation and digital capability were positively associated with digital transformation, and that transformation was the path through which those conditions related to revenue and the business model [18]. The measured outcome in that survey is progress, not failure. These studies are not columns in a public venture file. A classifier trained on closure can rank firms that shut down. It cannot tell a founder that the current transformation is failing because data maturity is behind the AI ambition, or because exploitation has pulled away from exploration.

Decision support for smaller firms has taken a different form. Organizational readiness for digital transformation is still framed as a construct to define, assess, and improve [19]. Co-word maps of digital transformation and innovation recover research themes [20]. They do not produce a firm-level score. Early-warning models exist where the label is a credit default [21]. A public venture file can record that default, a closure, or a funding round. It does not record digital-transformation failure.

Knowledge rules are inspectable and brittle. Data-driven models absorb residual structure and do not say which rule fired [22]. Certainty factors let

a rule carry a degree of belief and let several positive rules combine without being treated as a full probability [23]. Fuzzy classifiers treat the weight on a rule as part of the specification [24]. Belief-rule bases carry grades of belief by evidential reasoning [25], and that idea has been used for behavior prediction [26] and for ensemble classification [27]. KBEWS keeps the positive-evidence product as the knowledge score and anchors it to a gradient-boosting probability [28]. Hybrids that place a rule base beside a learned model have been built for trading policies [29] and, in this journal, for environmental early warning [30]. Neither target is a venture's transformation outcome, and neither uses the learned model to calibrate a certainty factor. The contribution is threefold: (i) a compact, editable rule base for digital-transformation failure; (ii) a certainty-factor anchor that returns a probability a decision maker can threshold; and (iii) a reproducible cohort benchmark that separates what the rules contribute from what the anchor contributes. Section 4 shows that the rules carry the ranking and that the anchor's distinct contribution is calibration.

2. Proposed KBEWS framework

2.1. Venture state

Each venture is a vector of twelve indicators on $[0, 1]$: digital leadership ℓ, digital capability c, financial slack s, data maturity m, transformation ambition a, ecosystem ties e, exploration x, exploitation i, AI ambition g, business-model adaptation b, firm age, and firm size. Twelve irrelevant noise

indicators are observed as well. They are available to the data model and invisible to the rules. Ten percent of the twelve signal values are missing. A rule whose input is missing abstains rather than being imputed.

2.2. Knowledge-rule layer

Seven rules convert the indicators into firing strengths in $[0, 1]$. The strengths were fixed before any cohort was scored. For a gap between a driving indicator and a lagging indicator the strength is a clipped linear excess. For a joint condition the strength is the minimum of the two excesses, so both conditions must hold before the rule contributes. Writing $\mathrm{clip}(u) = \min(1, \max(0, u))$,

$$
\begin{aligned}
s_1 &= \mathrm{clip}((a - c - 0.12)/0.50), \quad & s_2 &= \mathrm{clip}((a - s - 0.15)/0.50), \\
s_3 &= \mathrm{clip}((g - m - 0.12)/0.50), \quad & s_4 &= \mathrm{clip}((|x - i| - 0.22)/0.45), \\
s_5 &= \mathrm{clip}\big(\min\big(\tfrac{a-0.58}{0.30}, \tfrac{0.40-e}{0.30}\big)\big), & & \\
s_6 &= \mathrm{clip}\big(\min\big(\tfrac{a-0.55}{0.30}, \tfrac{0.42-\ell}{0.32}\big)\big), & & \\
s_7 &= \mathrm{clip}\big(\min\big(\tfrac{a-0.58}{0.30}, \tfrac{0.38-b}{0.30}\big)\big). & &
\end{aligned}
\tag{1}
$$

The seven mechanisms are capability gap, resource misfit, data deficit, ambidexterity imbalance, ecosystem isolation, leadership void, and weak adaptation. Their certainty weights are $w = (0.78, 0.70, 0.76, 0.58, 0.52, 0.64, 0.48)$. Positive evidence is combined by the parallel rule of Shortliffe and Buchanan

[23],

$$\mathrm{CF} = 1 - \prod_{k=1}^{7} \bigl(1 - w_k \tilde{s}_k\bigr), \tag{2}$$

where $\tilde{s}_k = 0$ if rule k abstains and $\tilde{s}_k = s_k$ otherwise, and each factor $w_k \tilde{s}_k$ is kept below 1. CF is a degree of belief that a failure mechanism is active. It is not yet a probability that can be thresholded.

2.3. Certainty-factor anchor

The data model is a histogram gradient-boosting tree of depth 3, learning rate 0.08, at most 160 iterations, minimum leaf size 20, and ℓ_2 penalty 0.1 [28]. It is fit to all twenty-four indicators, with missing values left missing. It returns a failure probability p_d. The anchor is a logistic regression on five quantities built from out-of-fold predictions, so the anchor does not see the test ventures while it is being fit:

$$\begin{aligned} \operatorname{logit} p = {} & b_0 + b_1 \operatorname{logit} p_d + b_2 \,\mathrm{CF} + b_3 \,\mathrm{CF}(1 - p_d) \\ & + b_4 s_{\max} + b_5 n_{50}. \end{aligned} \tag{3}$$

Here $s_{\max}$ is the strongest firing and n_{50} is the number of rules with strength at least 0.50. The product $\mathrm{CF}(1 - p_d)$ is available to raise the warning when the data model is silent. Whether the fitted anchor uses it that way is an empirical result, reported in Section 4, not an assumption. After the anchor is fit, the tree is refit on the full training split and Eq. (3) is applied to the held-out ventures.

Algorithm 1. KBEWS warning.

Input: venture indicators; the seven rules and weights in Eq. (1).

Output: warning probability p and the fired-rule trace.

1. Evaluate $s_1, \ldots, s_7$. A rule with a missing input abstains.
2. Combine the fired rules into CF by Eq. (2).
3. On the training split, build out-of-fold boosting probabilities with three stratified folds.
4. Fit Eq. (3) on those out-of-fold probabilities, CF, and the rule summary.
5. Refit the boosting model on the full training split and apply the anchor to the held-out ventures.
6. Return p and every rule whose strength is at least 0.50.

KBEWS differs from a data-only classifier in two structural ways. The knowledge layer is a separate symbolic score, not a set of features buried inside the tree. The anchor then makes that score numerically usable, and the fired rules remain visible after the probability is issued. Figure 1 shows one held-out failure in which that separation matters.

3. Experimental design

No public venture file observes digital-transformation failure, so the experiment uses a synthetic population whose label is defined by the mechanisms the rules approximate. Each cohort contains 500 ventures drawn with

its own seed. Latent leadership, capability, slack, data maturity, ambition, ecosystem ties, exploration, and exploitation are Gaussian and then clipped to $[0, 1]$. AI ambition is a noisy mixture of ambition. Adaptation, age, and size are drawn independently. The failure log-odds are a stack of threshold events: a three-way corner of high ambition, low capability, and low data maturity; ambition ahead of slack; AI ambition ahead of data maturity; an exploration–exploitation gap; ambition with weak ties; ambition without digital leadership; and a protective effect when adaptation is high. A size effect under high ambition is included and is deliberately absent from Eq. (1), so the data model has a residual signal the rules cannot see. Gaussian noise is added on the log-odds scale. Observed indicators equal the latents plus measurement noise, and 10% of signal cells are deleted. The mean failure rate across cohorts is 0.484. That rate describes firms already pursuing a transformation. It is not an estimate for all new firms.

Eighty cohorts give eighty matched comparisons. On each cohort a single stratified 70/30 split is drawn. Four systems are fit on that split and scored on the same held-out ventures: logistic regression on median-imputed, standardized indicators (neither the rule layer nor the nonlinear model); gradient boosting (nonlinear model, no rules); the certainty factor alone (rules, no data model); and KBEWS (both). Because each cohort is an independent draw, the paired Wilcoxon signed-rank test is the appropriate comparison of two systems across cohorts [31]. Tests are two-sided. The primary ranking metric is the area under the precision–recall curve [32]. The primary

calibration metric is the Brier score [33]. Recall at a false-positive rate of 0.10 is the early-warning operating point. The implementation is scikit-learn [34]. Fixed seeds make every cohort, split, and figure regenerable from the accompanying program.

4. Results and discussion

Table 1: Mean performance over 80 independent cohorts. Lower Brier scores are better. Higher values are better for AUPRC, AUROC, and recall.

Method	AUPRC	AUROC	Brier	Recall at FPR 0.10
Logistic regression	0.630	0.630	0.244	0.240
Gradient boosting	0.628	0.623	0.265	0.246
Knowledge base	0.681	0.672	0.270	0.316
KBEWS	0.683	0.671	0.226	0.318

Table 1 is the 2×2 reading of the system. Removing both the rule layer and the nonlinear model leaves logistic regression, with AUPRC 0.630 and Brier score 0.244. Replacing the linear model with gradient boosting does not help: AUPRC is 0.628 and the Brier score worsens to 0.265. The label is a set of thresholds, and the twelve noise indicators give the tree something to overfit. Putting the rules in and taking the data model out raises AUPRC to 0.681 and recall at a 10% false-positive rate from 0.246 to 0.316. That is the knowledge-layer gain. Enabling both components, which is KBEWS,

leaves AUPRC essentially unchanged at 0.683 and cuts the Brier score from 0.270 to 0.226.

Table 2: Paired Wilcoxon tests of KBEWS against each of the other three systems, using the 80 matched cohorts.

Comparison	Metric	Baseline	KBEWS	p-value
KBEWS vs. gradient boosting	AUPRC	0.628	0.683	7.74×10^{-13}
KBEWS vs. gradient boosting	Brier	0.265	0.226	8.15×10^{-15}
KBEWS vs. logistic regression	AUPRC	0.630	0.683	2.35×10^{-12}
KBEWS vs. logistic regression	Brier	0.244	0.226	2.41×10^{-13}
KBEWS vs. knowledge base	AUPRC	0.681	0.683	0.385
KBEWS vs. knowledge base	Brier	0.270	0.226	7.85×10^{-15}

Table 2 separates the two claims. Against either data-only model, both ranking and calibration favor KBEWS, and the paired p-values are far below 0.001. Against the knowledge base alone, the AUPRC comparison is not significant ($p = 0.385$). The Brier comparison is. Recall at a false-positive rate of 0.10 tells the same story: 0.318 for KBEWS, 0.316 for the knowledge base ($p = 0.855$), 0.246 for gradient boosting ($p = 8.19 \times 10^{-9}$), and 0.240 for logistic regression ($p = 3.66 \times 10^{-10}$). A deployment that only needs a ranking can publish the certainty factor. A deployment that needs a threshold needs the anchor, because a certainty factor of 0.60 is not a 60% probability.

Mean anchor coefficients across the 80 cohorts are $b_0 = -0.57$, $b_1 = 0.12$, $b_2 = 0.69$, $b_3 = -0.12$, $b_4 = 0.25$, and $b_5 = 0.39$. The certainty factor and

the count of strong rules dominate. The weight on $\operatorname{logit} p_d$ is small. The disagreement product $\mathrm{CF}(1 - p_d)$ receives a small negative weight, so the fitted anchor does not implement a rule of the form "raise the warning only when the data model is silent." It treats CF as the level of the warning and uses the data model as a weak adjustment. That is why KBEWS and the knowledge base rank ventures almost equally, while their probabilities do not.

Figure 1 makes the trace concrete on one held-out failure. Data maturity is 0.125 and AI ambition is 0.622, so rule R3 (data deficit) fires at strength 0.754. Exploration is 0.480 and exploitation is 0.978, so rule R4 (ambidexterity imbalance) fires at strength 0.617. Firm age is unobserved and does not enter a rule. The gradient-boosting model assigns probability 0.042. The certainty factor is 0.809. KBEWS assigns 0.785. A threshold of 0.5 would have cleared this venture on the data model and flagged it on KBEWS. The panel is one audited case, not a performance summary.

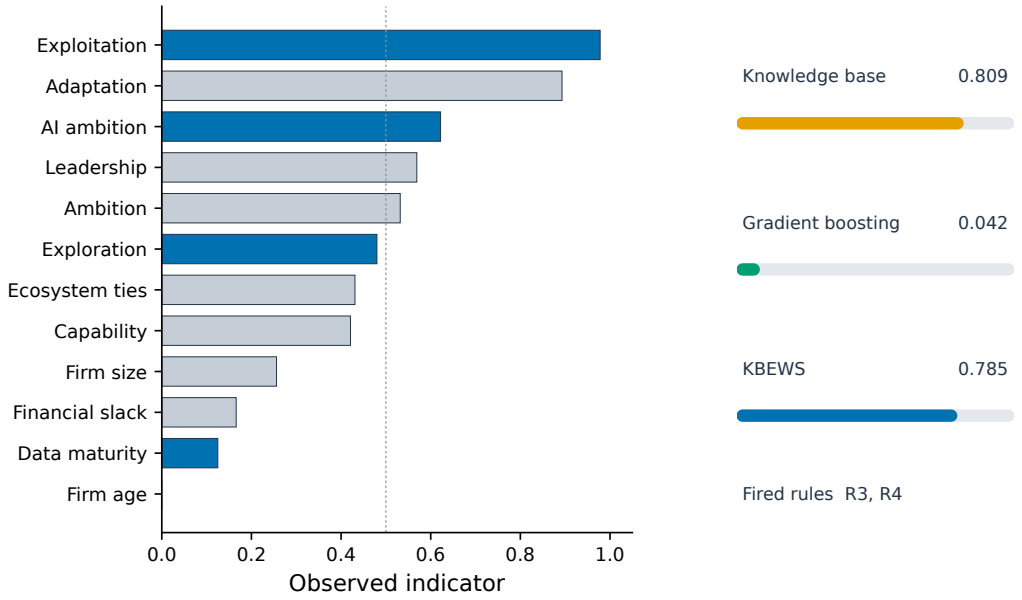


Figure 1: One held-out failure. Blue bars are inputs of rules that fired (R3, data deficit; R4, ambidexterity imbalance). Gray bars are observed indicators that did not enter a fired rule. Firm age was missing. The right-hand strips are the knowledge score, the data-only probability, and the KBEWS probability for the same venture.

Figure 2 shows why the Brier result matters. Predicted probabilities from all 12,000 held-out venture-rows are grouped into equal-width bins, and the vertical axis is the observed failure frequency in each bin. The knowledge score lies above the diagonal through the middle of the range: a moderate certainty factor overstates the failure frequency, and the score rarely approaches zero. Gradient boosting lies below the diagonal once the predicted probability is high, so a large p_d understates the failure frequency. KBEWS tracks the diagonal from about 0.4 upward. The anchor does not

create a new ranking. It converts the rule score into a probability whose stated value is closer to the frequency a decision maker will actually observe.

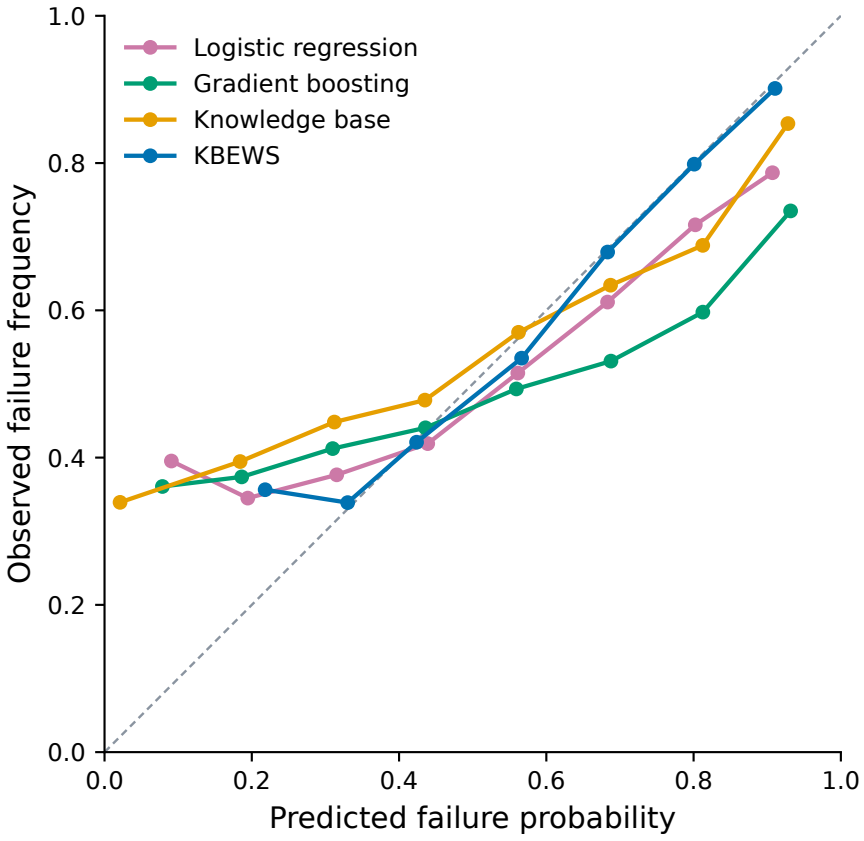


Figure 2: Reliability of the four scores on the pooled held-out rows (n = 12,000). The dashed line is perfect calibration. A point is the mean predicted probability in an equal-width bin against the observed failure frequency in that bin. Bins with fewer than 40 rows are omitted.

The studies cited in the introduction name capabilities, leadership, analytics inertia, and ecosystems as conditions of a transformation. They do not rank ventures, and Table 2 does not replicate them. The table compares

four scores on a mechanism-labeled draw.

The result places KBEWS inside knowledge-based decision support rather than inside black-box prediction. The venture is flagged because named rules fired and the anchored probability crossed a threshold the user can set. Explainability comes from Eq. (1) and Eq. (2), not from a post-hoc importance method applied after the tree has hidden the mechanism.

5. Limitations

The benchmark contains no person-level attributes, so sex and gender were neither measured nor analyzed. The stronger limit is external validity. Indicators, missingness, noise covariates, and the failure label are generated. Nothing in Table 1 estimates how often real entrepreneurial firms fail at digital transformation, and the rule weights are a specification, not a record of what founders believe. The anchor also does not improve ranking over the rules; a paper that claimed an accuracy gain over the knowledge base would be claiming a result this design does not support. Replacing the generator with a file that uses the same columns and an observed failure label does not require a change to the rules or to Eq. (3). Until that file exists, the numbers justify the system. They do not justify an industry claim.

6. Conclusion

This Short Communication presented KBEWS, a knowledge-based early-warning system for digital-transformation failure in entrepreneurial firms.

Seven graded rules produce a certainty factor. A gradient-boosting model produces a data-driven probability. A logistic anchor returns one probability and leaves the fired rules visible. Across 80 independent cohorts the rules, not the data model, carry the ranking. The anchor's contribution is calibration: the Brier score falls from 0.270 to 0.226, and the reliability curve moves onto the diagonal, while AUPRC is statistically unchanged from the knowledge base alone. KBEWS is a decision-support layer. Its output is a warning a person can threshold and a mechanism a person can inspect.

CRediT authorship contribution statement

Hemasri Babburu: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review and editing.

Declaration of competing interest

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data used in this study are synthetic and reproducible. `kbews.py`, with seed 23, generates the cohorts, the paired tests, Figure 1, and Figure 2. No third-party microdata and no personal data were used.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Cursor (Grok 4.7) for language drafting, reference checking, and to implement the experiment code. The tables and figures were generated by that code. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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